4.4 Writing Equations of Parallel and Perpendicular Lines

Name _____ Period ____ Date _____

Algebra I 4.4 · Student sheet

I can…

- identify parallel and perpendicular lines from their slopes.
- write equations of parallel lines.
- write equations of perpendicular lines.

Vocabulary · fill in the missing word

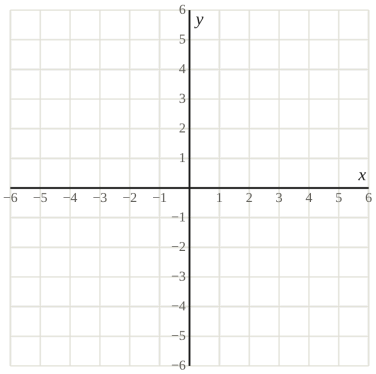
Parallel lines — lines in the same _____ that never meet; nonvertical parallel lines have equal slopes

Perpendicular lines — lines that meet at right _____; their slopes are negative reciprocals (product −1)

Negative reciprocal — flip the _____ and change the sign: $\frac{2}{3} \rightarrow -\frac{3}{2}$

Part 1 · Square corners

Graph y = 2x + 1 and y = −½x + 3. Check the angle with the corner of an index card. Then graph y = 2x − 3.



What do you notice about the slopes 2 and −½? _____

How is y = 2x − 3 related to y = 2x + 1? _____

Part 2 · Notes: fill in the blanks

Parallel lines have _____ slopes.

Perpendicular slopes are _____ _____; their product is _____.

Perpendicular to slope ¾: _____.

Part 3 · Examples (work them with the class)

Example 1

Which lines are parallel? Which are perpendicular?

a: y = 3x − 2 b: $y = -\frac{1}{3}x + 5$ c: 6x − 2y = 8

Example 2

Write an equation of the line through $(2, 5)$ parallel to $y = -3x + 1$.

Example 3

Write an equation of the line through $(-4, 1)$ perpendicular to $y = 2x - 7$.

Part 4 · Partner practice

Partner 1

Write the line through $(0, -2)$ parallel to $2x + y = 6$.

Partner 2

Write the line through $(3, 3)$ perpendicular to $y = \frac{3}{4}x$.